

Git and GitHub Workflow

Prerequisites

Before starting, install Git and create a GitHub account.

1. Install Git

Download and install the Git desktop application from:

<https://git-scm.com/downloads>

2. Create a GitHub Account

Sign up for a GitHub account:

<https://github.com/signup>

Git Workflow

The basic Git workflow consists of the following steps:

1. Initialize a local Git repository.
 2. Configure Git user information.
 3. Stage files for commit.
 4. Commit changes locally.
 5. Connect the local repository to a remote repository.
 6. Push changes to the remote repository.
 7. Pull the latest changes from the remote repository.
 8. Delete the local repository (optional).
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1. Create a New Local Git Repository

Navigate to your project folder and initialize a Git repository:

```
git init
```

This creates a `.git` folder and converts the project directory into a Git repository.

2. Configure Git User Information

Configure your name and email address. This is typically a one-time setup.

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

To verify the configuration:

```
git config --global --list
```

3. Add Files to the Staging Area

Use `git add` to stage files that you want to include in the next commit.

Add all files and folders

```
git add .
```

or

```
git add -A
```

Add a specific file

```
git add filename
```

Example:

```
git add package.json
```

Add files with a specific extension

```
git add "*.ts"
```

Add all files from a specific folder

```
git add foldername/
```

Check the staging status

```
git status
```

4. Commit Changes to the Local Repository

Commit the staged changes with a descriptive commit message:

```
git commit -m "Add initial Playwright framework"
```

To view the commit history:

```
git log --oneline
```

5. Connect the Local Repository to GitHub

Create a new repository on GitHub, then connect your local repository to it.

```
git remote add origin https://github.com/USERNAME/REPOSITORY.git
```

Example:

```
git remote add origin https://github.com/pavanoltraining/myproject.git
```

Verify the remote repository:

```
git remote -v
```

Note: Replace `USERNAME` and `REPOSITORY` with your GitHub username and repository name.

6. Push Changes to GitHub

First, check your current branch:

```
git branch
```

If your main branch is `main`, push using:

```
git push -u origin main
```

For a repository using `master`:

```
git push -u origin master
```

After the first push, you can usually use:

```
git push
```

GitHub Authentication

GitHub no longer accepts a normal account password for Git operations over HTTPS.

When authentication is required, use a **GitHub Personal Access Token (PAT)** instead of your GitHub password.

7. Pull the Latest Changes from GitHub

Use `git pull` to download and merge the latest changes from the remote repository.

Option 1: Fetch and merge manually

```
git fetch origin  
git merge origin/main
```

Option 2: Fetch and merge automatically

```
git pull origin main
```

For a `master` branch:

```
git pull origin master
```

Recommended workflow

Before starting new work, pull the latest changes:

```
git pull
```

8. Delete the Local Git Repository — Optional

If you want to remove Git tracking from a project while keeping the project files, delete the `.git` folder.

Windows PowerShell

```
Remove-Item -Recurse -Force .git
```

Windows Command Prompt

```
rmdir /s /q .git
```

macOS/Linux

```
rm -rf .git
```

Warning: Deleting the `.git` folder removes the local Git history and repository configuration. Your project files will remain, but the folder will no longer be a Git repository.

Quick Git Workflow

The most commonly used commands are:

```
git init

git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"


git status
git add .
git commit -m "Initial commit"


git remote add origin https://github.com/USERNAME/REPOSITORY.git
git push -u origin main


git pull
```

Typical Day-to-Day Workflow

```
git pull

git status
git add .
git commit -m "Update test cases"
git push
```